



UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The normal phase
4. The antiferromagnetic phase
5. The superconducting phase
6. Conclusions and outlook

Cuprates and Hubbard physics

Nobel Prize in Physics 1987



Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2



Photo from the Nobel Foundation archive.

K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba–La–Cu–O System

J.G. Bednorz and K.A. Müller

IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

What makes cuprates interesting?

- ◇ Superconductors that are insulating at normal state;
- ◇ Rich phase diagrams;
- ◇ Unclear origin of Cooper pairing;
- ◇ High- T_c superconductivity.

The timeline of high- T_c superconductivity

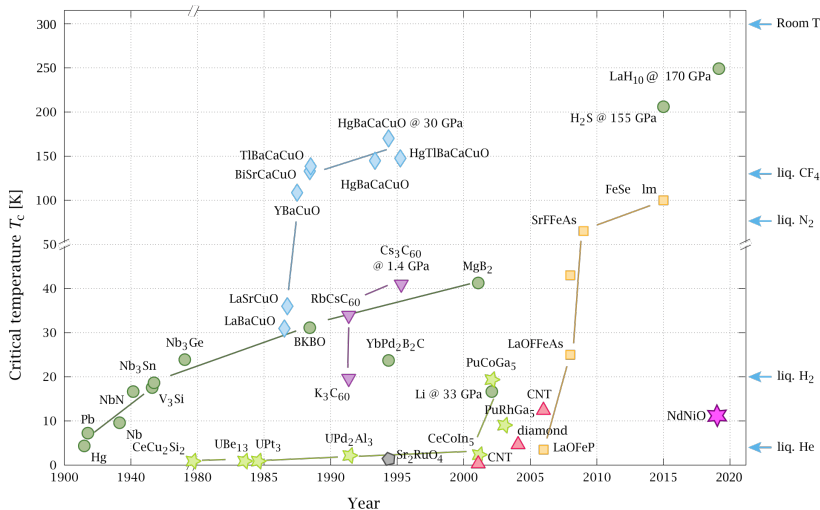


Figure 2: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](https://creativecommons.org/licenses/by-sa/4.0/).

The timeline of high- T_c superconductivity

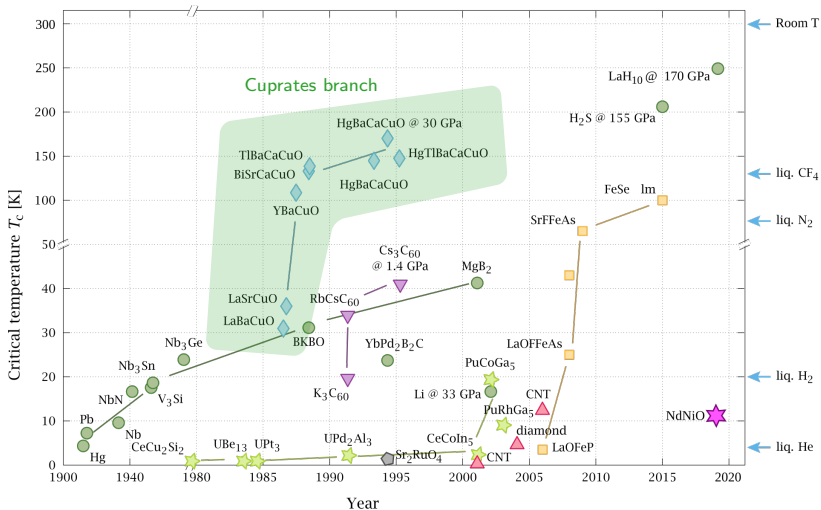


Figure 2: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](#).

The phase diagram of cuprates

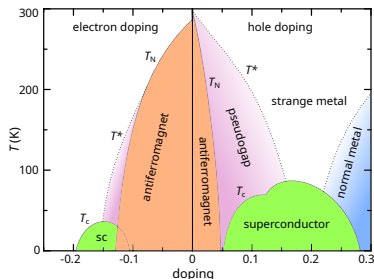
Cuprates are the prototypical example of **strongly correlated materials**:

- ◇ **Antiferromagnetism** at null and low doping;
- ◇ ***d*-wave superconducting** domes.

Both arise from **strong electronic correlations**.



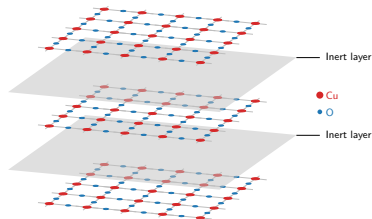
Failure of band theory in describing the insulation mechanism.



Schematic phase diagram of cuprates. Author: Holger Motzkau, [Wikimedia commons](#), CC BY-SA 3.0.

These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.



The single-band Hubbard model on the square lattice

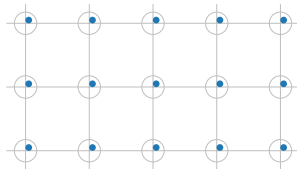
$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

What is the origin of the insulation mechanism?

The strongly coupled Hubbard model,

$$U \gg t,$$

can describe **Mott insulation**.

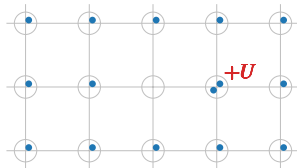


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Mott localisation: a many-body insulator

What is the origin of the insulation mechanism?

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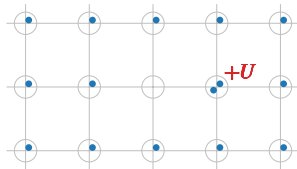
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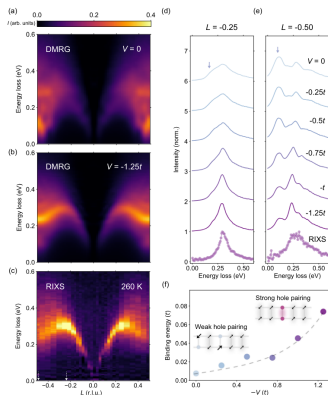


Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.



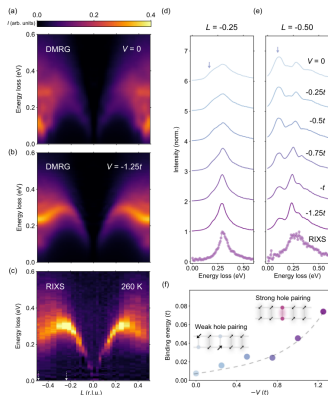
Recent developments: the extended Hubbard model



Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in certain cuprates.

Recent developments: the extended Hubbard model



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Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in certain cuprates.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

$\hat{H}_V$ can act as a source of **d -wave superconductivity**.

The model (EHM):

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

The motivation: compatibility with some aspects of cuprates physics, including d -wave superconductivity.

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

The method: theoretical and numerical Hartree-Fock method, at low temperature.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0 .

Variational principle:

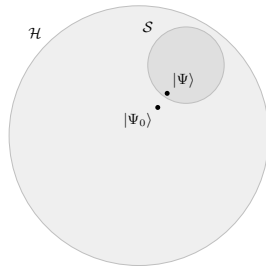
$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[\Phi] \geq E_0.$$

$\Downarrow$

Constrained minimisation procedure

$$|\Psi\rangle : \left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

The Hartree-Fock subspace of states

$\mathcal{S}$ is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

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with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Each gaussian state can be identified **uniquely** by fixing

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

These are the **variational parameters** collected in the vector $\mathbf{v}$. We aim to find the optimal $\mathbf{v}$, such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle.$$

Phase transition: spontaneous breaking of some symmetries of the model.

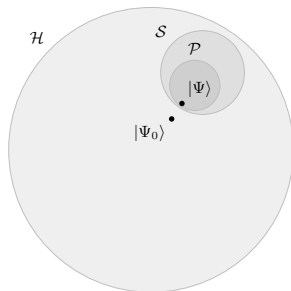
Phase X : shows symmetries $Y, Z \dots$



Restrict to $\mathcal{P} \subset \mathcal{S}$, gaussian states with symmetries $Y, Z \dots$



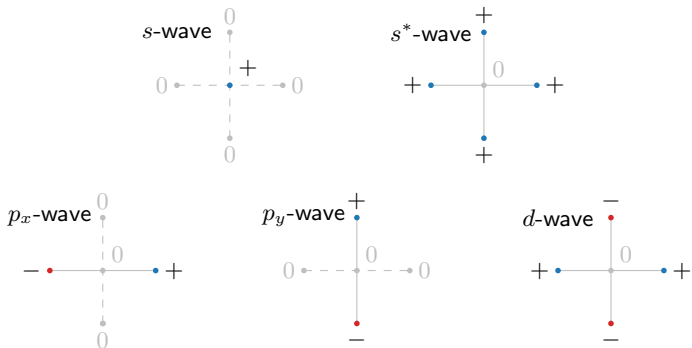
Symmetry selection rules on $\mathbf{v}$ entries.



Identify symmetries $\rightarrow$ Derive $\mathbf{v}$ $\rightarrow$ Solve the variational problem.

The spatial symmetry channels

It is useful to introduce five spatial symmetry channels:



Function f on NNs: decomposition in harmonics $f^{(\gamma)}$.

The normal phase

The variational problem for the normal phase

Normal phase: no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[u^{(s^*)} \right],$$

where $u^{(s^*)}$ is the s^* -wave component of $u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$.

$\Downarrow$

Renormalisation of kinetics

Hopping amplitude is **isotropically renormalised**,

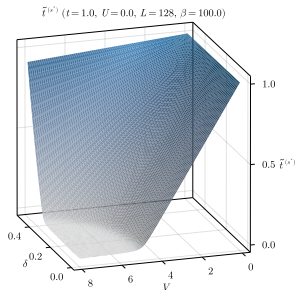
$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2},$$

with the **theoretical constraint**:

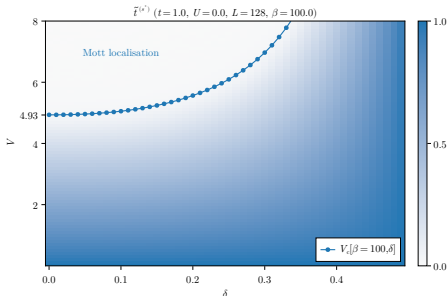
$$0 < \tilde{t}^{(s^*)} < t \quad \implies \quad \text{Possible Mott localisation.}$$

Complete flattening of bare bands

In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_c[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$.



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised and **bands are flattened**.

The role of V/t :

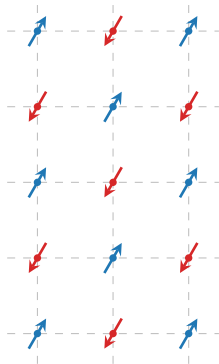
- ◇ **Renormalise hopping;**
- ◇ **Induce complete Mott localisation:**

Effective kinetics energy scale $\ll$ Interaction energy scale .

The role of U/t :

- ◇ Nothing.

The antiferromagnetic phase



In cuprates, the AF phase is **commensurate**:

- ◇ 2-sites periodic in each direction;
- ◇ Sublattices oppositely magnetised;
- ◇ Zero net magnetisation.

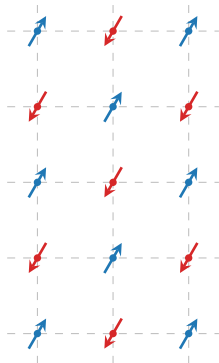


Staggered magnetisation m :

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with i a site.

Commensurate antiferromagnetism in cuprates



In cuprates, the AF phase is **commensurate**:

- ◇ 2-sites periodic in each direction;
- ◇ Sublattices oppositely magnetised;
- ◇ Zero net magnetisation.



Staggered magnetisation m :

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with i a site.

Half-filling restriction

We limit to $\delta = 0$ (half-filling) due to **stability considerations**.

Variational parameters for the AF phase:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix},$$

with $u^{(s^*)}$ defined as for the normal phase.

Renormalisation of kinetics

Mott localisation effect **formally** identical to the normal phase:

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2},$$

with the **theoretical constraint**: $0 < \tilde{t}^{(s^*)} < t$.

The cooperation on magnetisation

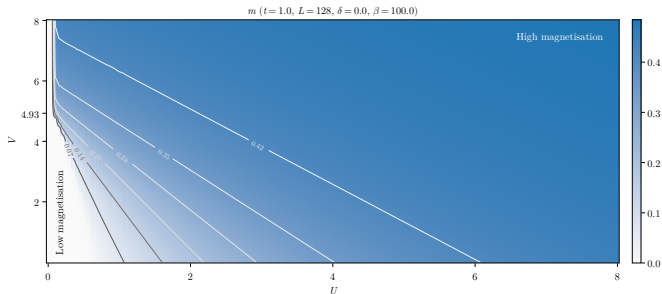


Figure 5: Numerical results for m . The gray curves indicate isovalued lines.

The cooperation on magnetisation

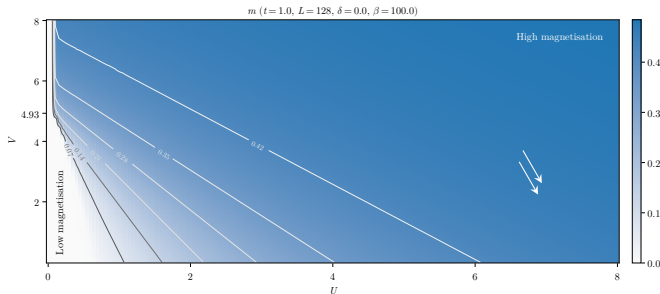


Figure 5: Numerical results for m . The gray curves indicate isovalued lines.

$$U \gg V, t$$

“Conventional” Hubbard magnetisation, induced by the ratio U/t .

The cooperation on magnetisation

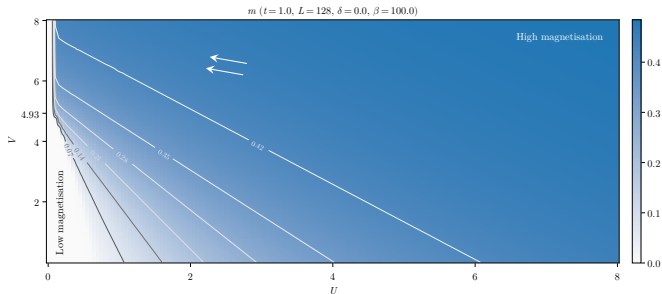
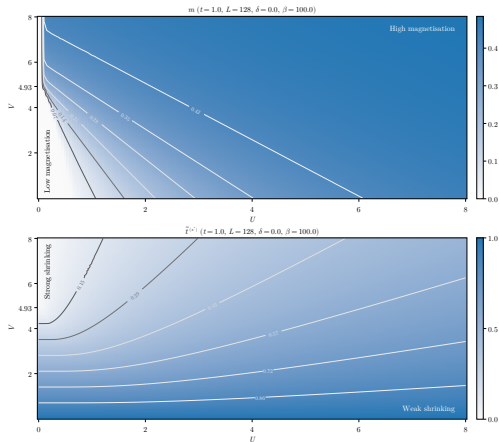


Figure 5: Numerical results for m . The gray curves indicate isovalued lines.

$$V \gg U, t$$

New “unconventional” source of magnetisation, induced by the ratio V/t .

The Mott enhancement of magnetisation



At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where $r = \tilde{t}^{(s^*)}/U$.

Figure 6: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate isovalues.

The Mott enhancement of magnetisation

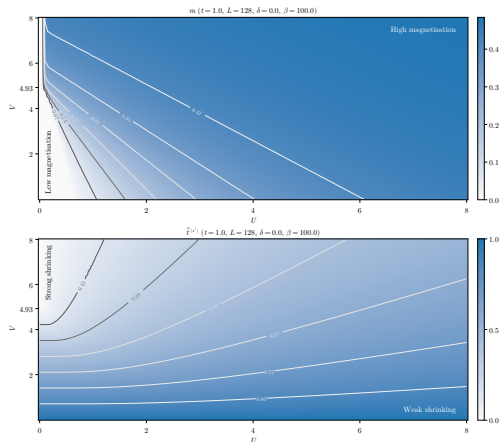


Figure 6: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate iso-valued lines.

At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where $r = \tilde{t}^{(s^*)}/U$.



To enhance m , either:

- ◇ Increase U ;
- ◇ **Decrease $\tilde{t}^{(s^*)}$** .

Effective hopping is:

- ◇ reduced by V ;
- ◇ **increased** by U .



The two interactions
“oppose” each other.

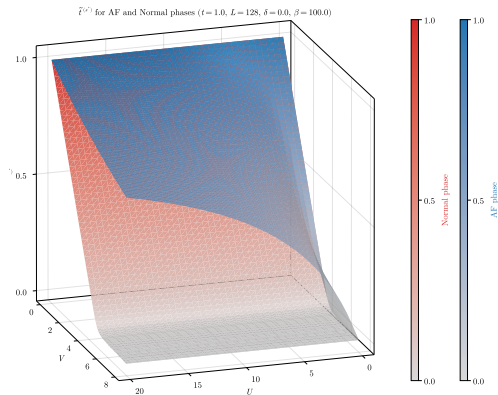


Figure 7: Comparative plot of $\tilde{t}^{(s^*)}$ for normal and AF phase.

The role of V/t :

- ◇ **Renormalise hopping;**
- ◇ **Magnetise the lattice using Mott physics;**

The role of U/t :

- ◇ Magnetise the lattice, conventional Hubbard physics.



Cooperation among interactions to magnetise the lattice.

The superconducting phase

SC phase: breaking of charge conservation.

Cooper pairing $\implies$ SC gap, roughly the “binding energy” of the pairs.

Gap symmetries and Cooper pairing channels

The gap is a function of momentum with harmonics $\tilde{\Delta}^{(\gamma)}$.

- ◇ **Singlet channel:** symmetric components,

$$\tilde{\Delta}^{(s)}, \quad \tilde{\Delta}^{(s^*)} \quad \text{and} \quad \tilde{\Delta}^{(d)}.$$

- ◇ **Triplet channel:** antisymmetric components,

$$\tilde{\Delta}^{(p_x)} \quad \text{and} \quad \tilde{\Delta}^{(p_y)}.$$

We limit to the **singlet channel**. Let the quantities:

$$u_{\mathbf{k}} \equiv \left\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right\rangle, \quad w_{\mathbf{k}} \equiv \left\langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \right\rangle.$$

Variational parameters: symmetric harmonics,

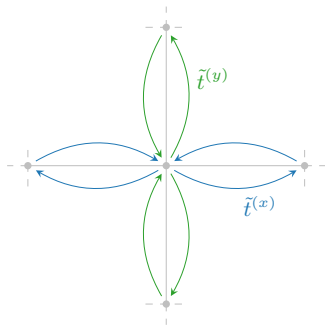
$$\mathbf{v} = \left(u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)} \right)$$

The role of each parameter

- ◇ $u^{(s^*)}, u^{(d)}$: renormalisation of kinetics;
- ◇ $w^{(s)}, w^{(s^*)}$ and $w^{(d)}$: components of the SC gap.

In the SC phase, the renormalisation of t gives two components,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}.$$



Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2},$$

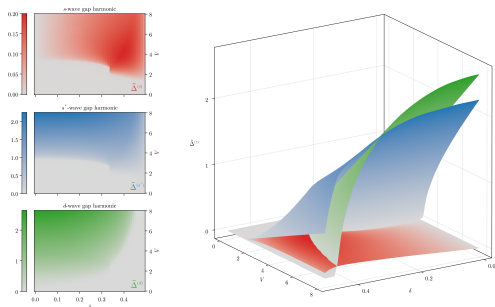
$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}.$$

$\Downarrow$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB.**

Gap symmetries in the singlet channel

Singlet gap $\tilde{\Delta}_s$ harmonics ($t = 1.0$, $U = 4.0$, $L = 128$, $\beta = 100.0$)



Gap components in
singlet channel:

$$\begin{aligned}\tilde{\Delta}^{(s)} &= -Uw^{(s)}, \\ \tilde{\Delta}^{(s^*)} &= Vw^{(s^*)}, \\ \tilde{\Delta}^{(d)} &= Vw^{(d)}.\end{aligned}$$

Figure 8: Singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$.

Internal constraint among singlet harmonics

$$V \sum_{\gamma \neq s} |w^{(\gamma)}|^2 \geq U |w^{(s)}|^2.$$

Symmetry mixing and insurgence of nematicity

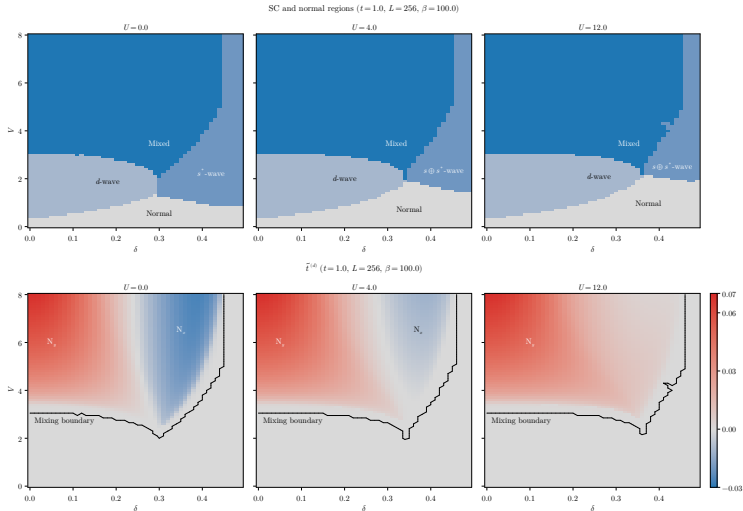
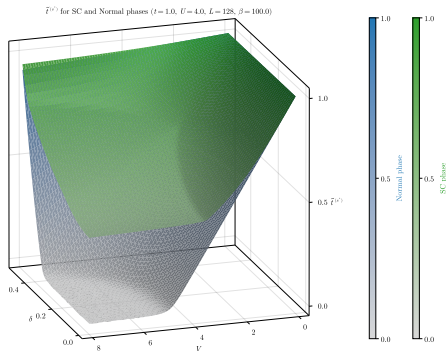


Figure 9: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.



Isotropic hopping
renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}.$$

Formally identical to
normal and AF phases.

Figure 10: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the SC phase.

The effective hopping is mainly isotropic:

$$|\tilde{t}^{(s^*)}| \gg |\tilde{t}^{(d)}|.$$

The role of V/t :

- ◇ **Renormalise hopping anisotropically;**
- ◇ **Induce mixed-symmetry superconductivity;**
- ◇ **Induce nematic symmetry breaking.**

The role of U/t :

- ◇ **Participate in superconductivity even if locally repulsive.**

Conclusions and outlook

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

At HF level, the non-local attraction:

- ◇ Acts as a source of Mott localisation;
- ◇ Enhances magnetisation at half filling;
- ◇ Can produce mixed-symmetry superconductivity;
- ◇ Can produce nematic superconductivity (unexpected, but observed!).

From here, we can follow various paths:

1. **Same model, other phases.**

Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism. . .

2. **Expand the model.**

Second nearest neighbours hopping, more refined potentials. . .

3. **More advanced numerical techniques.**

DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U . . .

or any linear combination of the three!

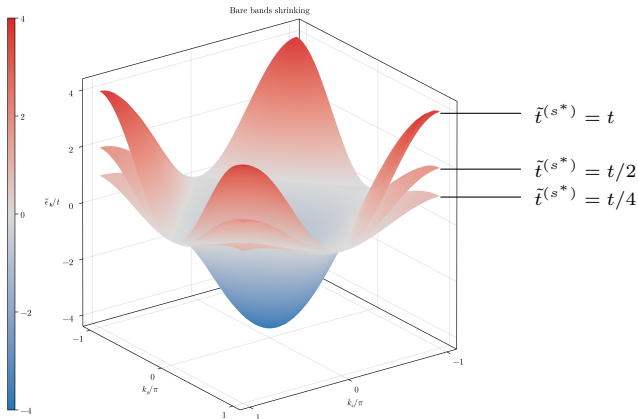
That's all, thank you!

Backup 1: details on Mott localisation

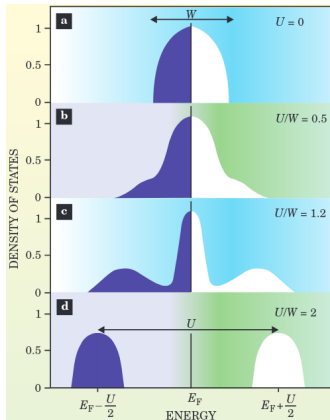
Bands flattening: a signature of Mott localisation (1/2)

Renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$:

$0 < \tilde{t}^{(s^*)} < t \implies$ **Flattening** of bands = Mott localisation.



Bands flattening: a signature of Mott localisation (2/2)



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

As we increase the ratio U/t , the many-body DoS undergoes a transition:

- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

Mott localisation: “electrons avoid being on the same site”. How does this relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Mott localisation at half-filling.



Singlet pairing $\leftrightarrow$ AF ordering.



HM at strong coupling:

$$\hat{H}_{\text{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E.$$

Superconductivity

Condensation of Cooper pairs with *d*-wave spatial structure.



Suppressed on-site pairing.

Backup 2: details on the HF method

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

The EHM hamiltonian is a quartic operator!

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$A(\mathbf{v}) = \mathbf{v},$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A ,

$$A(\mathbf{v}) = \mathbf{v}.$$



Strategy: “follow” the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i.$$

with $i > 0$, $g \in [0, 1]$.

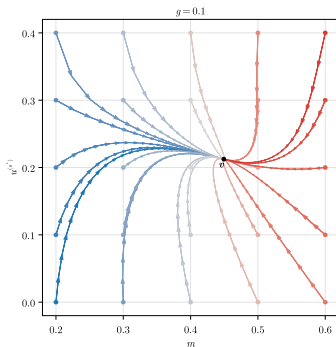


Convergence criterion:

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j,$$

with $\Delta \mathbf{v}$ the vector of tolerances.

Example path (AF) ($t = 1.0$, $U = 3.0$, $V = 6.0$, $L = 128$, $\beta = 100.0$, $\delta = 0.0$)



Example path for the AF phase.